

8 Structural Bottleneck Score

Ordered eigenvalues of the representation form the covariance matrix. PCA Score is defined as the fraction of dominant direction over top $[k]$ principal components.

Intuition: if the eigenvalue spectrum decays rapidly — a few large eigenvalues, then a long tail of small ones — the representation is low-rank and the PCA Score is high. Features are concentrated, neurons are monosemantic. If the spectrum is flat, variance is spread uniformly, the PCA Score is low, and we have high superposition.

Effective rank of a representation how many independent directions carry significant variance.

We decide when a neuron has become an information bottleneck and must be expanded through the combination of effective rank reduction and compression increase.

Structural bottleneck score captures a single dimensionless measure. Because both terms are normalized, score is independent of the feature dimensionality, and can be compared across layers and architectures. Identity mapping produces a score of zero, since neither compression nor structural reduction has occurred. The measure provides principled signal for architectural growth.

In this way, the Structural Bottleneck Score is not only a measurement but also a decision criterion which links information geometry directly to architectural adaptation. It enables network growth to be driven by representational structure rather than by optimization.

One important caveat: Score is a proxy, not a complete theory of interpretability. It captures the superposition dimension. Complementary metrics for semantic breadth, circuit modularity, and causal faithfulness remain open problems — I will return to this in the conclusion.

9 Training Dynamics

How do we build networks that are monosemantic by design? The answer comes in two parts: an active learning loop that shapes the training trajectory, and an adaptive growth mechanism that spawns new neurons when needed.

Active machine learning loop on the left is a standard pool-based procedure — iteratively selecting the most informative samples using acquisition functions based on predictive uncertainty, expected model change, diversity, and representativeness. The key insight is that this creates an optimal learning trajectory: the network never wastes capacity learning redundant representations.

The decision boundary panels at the bottom show what this means empirically: the network learns complex, non-linear boundaries while maintaining structured internal representations. The capacity goes exactly where it is needed — not uniformly across the network. Reinforcing synaptic connections upon correlated activity's been a Trait of both natural and artificial neural networks. Spatial organization of neurons close to the core function as well as the generation of a layered set of predefined types of New neurons's been a quality of natural neural networks only.

The biological inspiration here is deliberate. Natural neural networks spawn and reinforce connections during learning. We are instantiating that principle in a way that keeps PCA Score high throughout training.

10 Framework

Active Machine Learning: Acquisition function selects samples that maximally reduce uncertainty and increase diversity. This shapes what I call the neural plate — the overall tree topology — before training is complete. It determines where future growth will happen.

Adaptive Tree Growth During Training prevents polysemanticity. New branches spawn when rank reduces and compression increases. Network grows dedicated capacity rather than forcing existing neurons to encode new features.

Mechanistic Interpretability of the network: Each branch of the mature tree implements a specialised, testable computation — shape detectors, texture detectors, edge and boundary features, colour and intensity, part configurations, high-level semantic concepts. These branches are modular, localisable, and auditable.

11 Neural Plate

The causal tree figure on the right shows a fully trained AdaptiveNet: 1042 nodes, maximum depth 33. Node colours encode class influence — green for input, red for positive class influence, blue for negative, grey for neutral. Edge colours encode information flow direction and strength.

The four properties on the left tell us what this structure means. Coarse to fine: representations near the root are general, representations at the leaves are fine-grained and local. This mirrors the hierarchical structure we observe in biological visual cortex.

Organic growth: each node in this tree was spawned in response to a specific representational failure. The topology is not engineered — it emerged from the data. This is important: it means the tree reflects the actual structure of the problem, not our prior assumptions about it.

Modular circuits: each branch is a testable computational unit. You can ablate it, patch it, and trace its influence on the output — the standard toolkit of mechanistic interpretability, but now applicable to the entire network systematically.

And high PCA Score throughout: because each branch handles a specific representational subspace, the covariance within each branch is low-rank. The global PCA Score remains high even as the network grows.

12 Literature

Before discussing future directions, it is good to place this work within the existing literature. Throughout this talk I have argued that adversarial robustness became a mature scientific discipline because it developed objective metrics, standardized benchmarks, and reproducible comparisons (became measurable). Our central claim is that mechanistic interpretability can follow the same path.

Our work introduces two complementary contributions that naturally reinforce one another. The proposed Structural Bottleneck Score S provides a quantitative proxy for feature superposition, guides architectural growth during training. These are two sides of the same geometric framework. Both are principled proxy quantities built on the hypothesis that feature superposition is a fundamental latent variable linking interpretability and adversarial robustness.

The first half of the timeline illustrates a remarkable progression in adaptive neural architectures. Cascade-Correlation demonstrated that networks could grow during learning by adding neurons where additional capacity was needed. Net2Net and Network Morphism showed that networks could be widened or deepened while preserving their learned function, making architectural expansion practical and efficient. Adaptive Structure Learning and later Splitting Steepest Descent further advanced this direction by allowing networks to expand automatically during optimization. These methods have significantly advanced adaptive learning, but they remain primarily optimization-driven. They ask where additional capacity will accelerate convergence or reduce training loss most effectively.

Our approach builds directly on these ideas while introducing a complementary perspective. Instead of asking where optimization benefits most, we ask where the representation itself requires additional capacity.

The second half of the timeline highlights the rapid progress of quantitative interpretability. Network Dissection, Identifying Concepts in Deep Networks, and the Broden dataset established rigorous methods for quantifying semantic interpretability through human-labelled concepts. Toy Models of Superposition provided an elegant theoretical explanation for why polysemantic neurons naturally emerge, while MIS demonstrated that interpretability can also be quantified automatically through behavioural consistency.

Our work as a whole complements these important advances by focusing on a different aspect of the problem. Rather than measuring semantic alignment or behavioural consistency, we measure the underlying feature superposition itself. Using effective-rank estimation together with adversarial sampling, we obtain layer-wise and network-wide estimates of representational complexity without requiring concept labels or manual annotation. Our work is complementary and fundamentally different.

The objective is not to replace existing interpretability methods, but to provide the kind of quantitative, architecture-agnostic metric that enables reproducible comparison, principled adaptive growth, and ultimately a measurable science of mechanistic interpretability.

13 Value Alignment

The value alignment problem of AI has two components, shown in the diagram. Inner alignment asks whether the system's internal representations, objectives, and decision processes are consistent with intended values. Outer alignment asks whether the system's behaviour and world impact align with the chosen normative theory.

These two components are mutually reinforcing. You cannot verify outer alignment without inspecting internal representations — that is, without mechanistic interpretability. And inner alignment is meaningless if the external behaviour does not reflect the internal values. For neural network to comprehend human values, understand the context, place values within context requires the addressing of inner and outer alignment in terms of objective specification and informational asymmetries. Transparent artificial neural networks and the value alignment problem are two sides of the same coin, the coin being any central artificial intelligence. This is key/focal to the future of artificial intelligence: Complete mechanistic interpretability, that is neural network is segmented into clearly causal mechanisms, and comprehensive values alignment

This is why the banner on this slide says: interpretability and alignment are not competing objectives. They are complementary requirements. PCA Score and the framework contribute to inner alignment — they make the internal representations readable, auditable, and comparable across architectures.

The vision is an AI system that can be trusted the way a medical specialist is trusted at the point of diagnosis: not because we have blind faith in the output, but because we can inspect, validate, and understand the reasoning pathway.

14 Health Domain

As artificial intelligence assumes increasingly central roles in medicine, science, and public decision-making, performance alone will no longer be sufficient. Systems must also be transparent, interpretable, and aligned with human values. Interpretability and alignment are not competing objectives—they are complementary requirements for trustworthy artificial intelligence. Developing quantitative measures of interpretability is therefore not simply a technical challenge; it is a prerequisite for building AI systems that deserve our trust.

The quote comes from the core argument of this research: if we expect AI to participate in high-stakes decisions — medical diagnosis, judicial support, policy recommendation — we cannot evaluate it on accuracy alone. We must understand how those decisions are internally represented.

Three domains, three different stakes. In medical diagnosis: a physician receiving an AI recommendation must be able to inspect it, challenge it, and override it. Without mechanistic transparency, the AI is a black box that may be right for the wrong reasons. Quantitative interpretability provides a route to regulatory-grade evidence of transparency.

In public decision-making: democratic accountability requires that automated decisions be auditable by non-specialists, regulators, and civil society. An interpretability score is the kind of instrument that makes that possible.

In scientific discovery: an AI that discovers a pattern in genomics or materials science must be able to show its reasoning. Interpretable representations are a prerequisite for extracting genuine scientific knowledge — not just predictions — from learned models.

If we expect artificial intelligence to participate in high-stakes decisions—such as medical diagnosis—we cannot rely solely on empirical performance. We must also understand how those decisions are represented internally. Interpretability and alignment are therefore not independent objectives; they are complementary requirements for trustworthy AI. Quantitative measures of interpretability may provide the missing bridge between performance and trust.

15 Implications

01 — Interpretability Benchmarks: The research enables the same kind of standardised leaderboard for mechanistic interpretability that RobustBench provides for adversarial robustness. This is the most direct implication.

02 — Cross-Architecture Comparison: because PCA Score is architecture-agnostic, we can compare ResNets, Transformers, and MIVAANN trees on the same scale. This has never been possible with probing classifiers or activation patching.

03 — Training-Time Monitoring: PCA Score can be computed at every checkpoint during training. If it drops, the network is developing polysemantic representations. This gives practitioners an early- warning signal and a principled stopping criterion.

04 — Alignment Verification: quantitative interpretability provides the instrument for verifying inner alignment. If we can read the representations, we can ask whether they encode the intended values.

05 — Regulatory-Grade Evidence: medical device regulators, financial supervisors, and legal authorities need evidence, not intuition. A published, reproducible interpretability benchmark is the kind of evidence they can evaluate.

06 — Open Problems: connecting PCA Score to causal circuit analysis; extending the metric to large language models; developing complementary metrics for semantic breadth; and the key regularisation question — does training with PCA Score as an explicit objective improve both interpretability and robustness simultaneously?

16 Conclusion

Let me close with the chain that runs through this entire talk. Adversarial robustness succeeded as a science because it became measurable and comparable. Mechanistic interpretability has not yet taken that step. we hypothesize that feature superposition is a central latent variable...” ”our results suggest that feature superposition provides a useful geometric bridge between these two fields.

PCA Score — the fraction of representation variance captured by the top principal components — is our proposed quantitative proxy for interpretability. It is architecture-agnostic, computable during training, and formally linked to adversarial robustness through the superposition relationship.

MIVAANN demonstrates that monosemantic representations can be achieved by design: active learning shapes the training trajectory, adaptive neuron spawning prevents polysemanticity, and the resulting tree structure is mechanistically interpretable at every branch.

The closing thesis in the box on the right: if successful adversarial examples exist in polysemantic manifolds, then measuring those manifolds establishes a geometric metric for mechanistic interpretability — the missing bridge between performance and trust.

Three open questions for discussion: How does PCA Score relate to causal circuit-level interpretability? Can analogous metrics be developed for large language models? And does training with PCA Score as a regulariser simultaneously improve robustness? I look forward to your thoughts. Thank you.